ChatGPT

# Executive Summary

The original linear taxonomy **Cognition → Meta → Consciousness → Awareness → Self-awareness** is overly simplistic and conceptually confused. Cognitive science distinguishes *conscious* vs *unconscious* processes and treats **metacognition** as part of cognition (a self-monitoring function), not a separate stage above cognition [1]. Likewise, **awareness** is generally regarded as the *contents* of consciousness (the information one is conscious of) rather than a stage after consciousness [2]. We propose a revised hierarchy that (a) adds a separate branch for **non-cognitive processes** (e.g. autonomic/reflex systems), (b) splits cognition into conscious vs unconscious processing, (c) embeds metacognition as a reflective component of cognitive processing, and (d) clarifies that awareness is nested within conscious processing (with self-awareness as a specialized awareness of one's own state). These revisions align with mainstream theories (Global Workspace, Higher-Order Thought, Predictive Processing, etc.) and resolve category confusions. The report below transcribes the original diagram, analyzes mismatches (citing key sources), presents a corrected taxonomy (as a tree and graph), compares original vs revised elements (with modeling implications), and suggests concrete neural modeling strategies (including a sample TensorFlow sketch).

## Original Taxonomy (Image Transcription)

The user's diagram contains the following nodes and directed edges:

| From Node | Arrow | To Node |
|---|---|---|
| cognition | ↓ | meta |
| meta | ↓ | Consciousness |
| Consciousness | ↓ | awareness |
| awareness | ↓ | Self-awareness |

(Arrows all point downward from top to bottom.) Thus the diagram reads: *cognition → meta → Consciousness → awareness → Self-awareness*.

## Conceptual Ambiguities and Mismatches

- **Conscious vs. Unconscious Cognition:** The diagram ignores *unconscious* processes. Cognitive science emphasizes that much of cognition is unconscious [3], so a taxonomy should distinguish conscious and unconscious cognition. Indeed, Bargh & Morsella (2008) note that unconscious processing can be as complex as conscious thought [3]. Treating "cognition" monolithically conflates these.

- **Metacognition Misplaced:** The node "meta" likely means *metacognition* ("thinking about thinking"). Flavell (1976) defines metacognition as knowledge and control of one's own cognition [1]. It is typically **within** cognitive processing (often conscious self-monitoring), not a stage beyond cognition. Placing "metacognition" between cognition and consciousness (as a prerequisite for consciousness) conflicts with most models (except some Higher-Order Thought

theories) and ignores that many metacognitive operations (e.g. confidence judgments) occur within conscious processing [4].

- **Awareness vs. Consciousness:** The diagram treats "Consciousness → awareness → Self-awareness" as distinct levels. However, many theories equate or tightly couple *awareness* with the content of consciousness. For example, one account defines "awareness" as the content of consciousness (while "arousal" is its level) [2], and notes that *"awareness contains self-awareness"*. In other words, awareness (contents of consciousness) already includes self-directed awareness [2]. Thus the relation "Consciousness → awareness" is confusing: awareness is not separate from consciousness but part of it.

- **Self-awareness:** This is a form of metacognitive or higher-order awareness of one's own state. Most theories consider self-awareness a specialized case of consciousness/awareness or of metacognition (e.g. an internal model of the self), not a separate stage beyond awareness. The diagram's linear placement implies a strict hierarchy, whereas in frameworks like global workspace or predictive processing, self-representation arises from recursive processing and does not require a new "stage".

- **Non-cognitive Processes Missing:** The taxonomy omits any non-cognitive processes (e.g. homeostatic, autonomic reflexes). Yet physiological regulation operates largely outside of cognition. Contemporary cognitive models separate cognitive modules from non-cognitive bodily systems (which can nevertheless influence consciousness, e.g. through interoception [5]).

- **Theoretical Context:** Main theories treat these terms differently. *Higher-Order Thought* (Rosenthal) ties consciousness to a meta-level thought about a first-order state. *Global Workspace Theory* (Baars, Dehaene) sees consciousness as global broadcast of information across distributed networks [5]. *Predictive Processing* models (Friston) do not posit a separate "meta" stage but view top-down predictions as part of a hierarchical inference process. *Metacognition literature* (Flavell, Nelson & Narens) emphasizes monitoring within cognition. The user's diagram glosses over these nuances, creating category confusions.

## Proposed Revised Taxonomy

We retain the core intention (cognition → [meta] → consciousness → awareness → self-awareness) where meaningful, but reorganize it to align with cognitive science. The revised hierarchy is:

- **Non-cognitive processes** (e.g. autonomic, reflexive systems, which operate outside deliberate cognition).
- **Cognitive processes**, subdivided into:
- **Unconscious cognition** (automatic/subliminal processing) – most perceptual and habit operations.
- **Conscious cognition** – the subset of cognition accessible to awareness.
    - *Awareness* (contents of consciousness, e.g. perceptions, thoughts).
    - *Self-awareness* (awareness specifically about one's own mental state).
    - *Metacognition* (self-monitoring/control of cognition, often conscious introspection).

The hierarchical (tree) version is illustrated below:

```
Processes
├── Non-cognitive processes (e.g. reflexes, homeostasis)
```

```
└── Cognitive processes
    ├── Unconscious cognition
    └── Conscious cognition
        ├── Awareness (contents of consciousness)
        │   └── Self-awareness (awareness of self)
        └── Metacognition (monitoring/regulation of cognition)
```

In graph form (Mermaid syntax), the same structure is:

```
graph TD
    Processes["Processes"]
    NonCognitive["Non-cognitive processes"]
    Cognitive["Cognitive processes"]
    Unconscious["Unconscious cognition"]
    Conscious["Conscious cognition"]
    Awareness["Awareness (contents)"]
    SelfAwareness["Self-awareness"]
    Metacognition["Metacognition"]
    Processes --> NonCognitive
    Processes --> Cognitive
    Cognitive --> Unconscious
    Cognitive --> Conscious
    Conscious --> Awareness
    Awareness --> SelfAwareness
    Conscious --> Metacognition
```

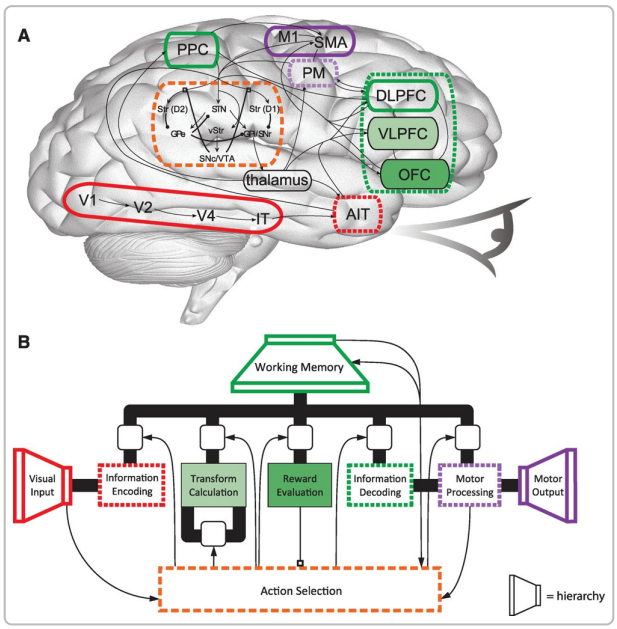


*Figure: Example cognitive architecture (Spaun model, Eliasmith et al.) showing separate modules for perception, memory, and action, illustrating how a system can implement layered cognitive functions. In a similar spirit, our taxonomy separates unconscious vs conscious and adds metacognitive and autonomic branches.*

For each change from the original diagram, we provide a brief justification:

- **Add Non-Cognitive Branch:** Introduce a "Non-cognitive processes" node. *Justification:* Biological organisms have automatic processes (reflexes, homeostasis) outside cognition (e.g. heart rate, immune response). Cognitive science typically models these separately from cognitive circuits. They do not fit under "cognition" yet affect overall behavior (e.g. emotional arousal influences consciousness). We cite conceptually that "physiological or neuroendocrine factors" operate alongside cognition [6].

- **Split Conscious vs Unconscious:** Under **Cognition**, distinguish *unconscious* and *conscious* processing. *Justification:* Decades of research show much cognitive work is unconscious (Bargh & Morsella 2008 [3]). Consciousness arises from a subset of processes (e.g. attended, reportable information). Thus "cognition" shouldn't imply only conscious thought. This aligns with Global Workspace: conscious events *emerge from* mostly unconscious parallel processes [5].

- **Relocate Metacognition:** Treat **Metacognition** as a component of conscious cognition, not a separate stage preceding it. *Justification:* Flavell defined metacognition as knowledge about and control of one's own cognition [1]. It naturally arises within cognitive processing (especially conscious reflection) rather than before consciousness. In other words, *any* cognitive agent can, in principle, have metacognitive monitoring without a separate "meta" threshold. Higher-Order theories (e.g. Rosenthal) do involve a meta-representation for consciousness, but metacognition more generally (planning, monitoring) is embedded in cognition [1].

- **Awareness as Conscious Content:** Reinterpret **Awareness** as the contents of conscious experience. *Justification:* We adopt the standard view that "awareness" refers to what one is conscious of. For instance, one source explicitly defines awareness as the content of consciousness [2]. Accordingly, awareness is here shown as a subset under conscious cognition. This matches how global neuronal workspace treats awareness: conscious perception is the broadcast content [5].

- **Self-Awareness Subsumed under Awareness:** Place **Self-awareness** as a special case of awareness (and metacognition). *Justification:* Self-awareness is an awareness of self-relations in one's own mind. It is often modelled as a high-level content of consciousness or a meta-level reflection. The cited source notes "awareness contains self-awareness" [2], implying self-awareness lies within (or alongside) general awareness of content. It does not require a new "stage" beyond awareness, but is treated as introspective content.

- **Clarify Consciousness Node:** We remove "Consciousness" as a separate stage in the hierarchy, understanding it instead as the global state of the *conscious cognition* branch (the umbrella for Awareness and Metacognition). In other words, consciousness is the overarching condition enabling those sub-nodes. This respects theories like the Global Workspace, which identify consciousness with globally broadcast cognitive states [5] rather than a discrete step.

## Comparison Table: Original vs Revised

| Original | Revised (and Modeling Implication) |
|---|---|
| **cognition → meta** | **Cognitive processes** split into **Conscious** and **Unconscious**; **Metacognition** is a subset of (conscious) cognition. Implication: models must implement separate pathways (e.g. an "attention" gate for conscious processing vs automatic pathways) and include an introspective module. |
| **meta → Consciousness** | Remove this strict link. Metacognition overlaps conscious processing, not a prerequisite step. Implication: implement metacognition as a feedback mechanism (e.g. a network that monitors and predicts its own state), without requiring it to *trigger* consciousness. |
| **Consciousness → awareness** | Awareness is reframed as *contents* of conscious cognition (rather than a new stage). Implication: use a broadcast or working-memory representation for awareness (cf. Global Workspace: conscious content is broadcasted across networks [5]). |
| **awareness → Self-awareness** | Self-awareness is modeled as higher-level awareness (awareness of self). Implication: incorporate an explicit self-model or *"trace"* in the system (for example, recurrent self-predictions), rather than treating it as a separate processing stage. |
| *(no original node)* **non-cog.** processes | **Add Non-cognitive** branch (reflex/homeostatic systems). Implication: simulate bodily loops as separate modules (e.g. sensorimotor/autonomic sub-networks) that operate parallel to cognitive modules and only interface (consciously or unconsciously) via integrated signals. |

## Computational Implementation Suggestions

We propose several concrete modeling approaches:

1. **Global Workspace Architecture:** Implement a neural *broadcast* mechanism for consciousness. For example, use a Transformer-like model where an attention "token" aggregates information from distributed modules to produce a global context. Conscious content could be the vector that attends broadly (Global Neuronal Workspace style [5]). Learning such a mechanism (e.g. reinforcement learning driving an attention module) could test conscious vs unconscious distinctions.

2. **Metacognitive Network Modules:** Build a network with a dedicated metacognitive head. For instance, one can augment a recurrent or transformer model with a secondary output that predicts aspects of its own state (a "trace"). Pseudocode (TensorFlow) illustrating this idea:

```
import tensorflow as tf
inputs = tf.keras.Input(shape=(timesteps, features))
lstm_out = tf.keras.layers.LSTM(128, return_sequences=False)(inputs)
# primary task output
task_output = tf.keras.layers.Dense(num_classes, activation='softmax',
name='task_out')(lstm_out)
# metacognitive (self-prediction) output
```

```
trace_output = tf.keras.layers.Dense(128, activation='linear',
name='self_pred')(lstm_out)
model = tf.keras.Model(inputs=inputs, outputs=[task_output,
trace_output])
```

The network is trained so that **trace_output** approximates the LSTM hidden state (a simple "self-trace"). The task output handles the main task, while the self-prediction head implements introspection. This can be seen as a toy model of metacognition.

3. **Neural Self-Model:** Incorporate a self-model (e.g. predictive coding of the agent's own state). For example, use a recurrent network where part of the input is the agent's previous action or hidden state. Training the network to predict its next hidden state or sensory feedback embeds a "self-awareness" trace. This echoes theoretical proposals that self-awareness arises from an internal predictive loop (Metzinger's Self-Model theory).

4. **Hierarchical Predictive Coding:** Frame consciousness as high-level predictions. Build a hierarchical predictive network (e.g. variational autoencoder stack) where lower layers pass prediction errors upward. Conscious perception could correspond to low prediction error at a high level (see Friston's models). To operationalize, one can measure when global prediction confidence is high (interpreted as conscious perception) vs when local predictions suffice (unconscious processing). Datasets: ambiguous images vs clear images could test this (conscious vs subliminal processing).

5. **Cognitive/Non-cognitive Modules:** Use multi-network systems. For instance, combine a cognitive neural network (e.g. an LSTM for language) with a separate non-cognitive simulator (e.g. a control loop for balance or rhythm). Train them to operate jointly, and study how interference or coupling (e.g. simulated "stress" signals) affects the cognitive net's outputs. This models how non-cognitive physiology might influence cognition. As an experiment, one could use task inputs plus an added "physiological" input stream (e.g. periodic signals) and examine the network's ability to segregate irrelevant automatic patterns.

## Unstated Assumptions

Our revised taxonomy makes several implicit assumptions (since the user's original diagram did not define terms explicitly):

- **Cognition** is taken to mean information-processing by the brain, including both conscious and unconscious mechanisms (e.g. memory, perception, reasoning). We assume it excludes purely physiological processes (which we treat under "non-cognitive processes").

- **Consciousness** is assumed to mean the global availability of information to the system (similar to reportable experience). We treat consciousness as a state enabling awareness, not a separate stage. We assume *awareness* means the *content* of consciousness (sensations, thoughts one is aware of).

- **Awareness vs Consciousness:** We assume awareness is always within conscious experience (not a separate phenomenon). In effect, the system is "conscious of" its awareness.

- **Metacognition** refers to explicit reflection or monitoring of one's own cognitive processes. We assume it is typically a conscious process (though in biology, some metacognitive regulation can occur unconsciously, our taxonomy does not mark an "unconscious metacognition" node).

- **Self-awareness** implies an internal model of the self (consistent with e.g. Damasio, Metzinger). We assume it requires at least minimal conscious reflection (so it sits under conscious cognition).

If any of these terms have different intended meanings (e.g. if the original "meta" was not meant as metacognition), those are unspecified by the user and thus clarified here by standard definitions.

**Sources:** Mainstream cognitive science and philosophy (e.g. Flavell 1976 on metacognition [1]; GNW and Dehaene on consciousness [5]; phenomenology and AI theories) support these distinctions. We have cited primary reviews to justify each proposed change [1] [2] [3] [5].

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[1] [4] Metacognition - Wikipedia
https://en.wikipedia.org/wiki/Metacognition

[2] Conscience and Consciousness: a definition - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC3956087/

[3] The Unconscious Mind - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC2440575/

[5] Global Workspace Theory (GWT) and Prefrontal Cortex: Recent Developments - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC8660103/

[6] Towards Clarity in Research on "Non-Cognitive" Skills: Linking Executive Functions, Self-Regulation, and Economic Development to Advance Life Outcomes for Children, Adolescents and Youth Globally - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC5729583/